Claim Normalization

Anish
2022075

Amartya Singh
2022062

Adarsh Kumar Jha
2022024

1 Introduction

This report presents our work on claim normalization, a natural language processing task that aims to convert informal social media posts into standardized, formal claims. We implemented and compared two transformer-based models: BART and T5. The evaluation demonstrates that BART outperforms T5 across all metrics, making it the preferred model for claim normalization tasks.

2 Preprocessing Steps

2.1 Text Preprocessing Pipeline

Our preprocessing pipeline consists of the following key components:

- **Contraction Expansion**: Converting shortened forms like "don't" to "do not" using the contractions library
- **Abbreviation Expansion**: Converting common abbreviations (e.g., "Gov." to "Governor") using a custom dictionary
- **URL and Handle Removal**: Eliminating URLs and Twitter handles using regular expressions
- **Hashtag Processing**: Converting hashtags to regular words by removing the '' symbol
- **Punctuation Normalization**: Retaining semantically important punctuation while removing duplicates
- **Whitespace Normalization**: Removing extra spaces and standardizing whitespace

2.2 Core Functions

- **expand_contractions()**: Expands contractions and abbreviations in text
- **clean_text()**: Removes URLs, handles, and normalizes text formatting
- **preprocess_text()**: Combines all preprocessing steps into one pipeline

3 Model Architecture and Hyperparameters

3.1 Models Overview

We implemented two sequence-to-sequence transformer models:

- **BART (Bidirectional and Auto-Regressive Transformers)**
  - Base architecture: facebook/bart-base
  - Input context: Social media post directly
  - Encoder-decoder architecture with bidirectional encoder and auto-regressive decoder
- **T5 (Text-to-Text Transfer Transformer)**
  - Base architecture: t5-small
  - Input context: Prefixed with "normalize:" + social media post
  - Unified text-to-text approach for all NLP tasks

3.2 Tokenization

- **BART Tokenization**: Direct tokenization without prefix
- **T5 Tokenization**: Task-specific prefix "normalize:" added to inputs
- Both used max length of 512 tokens for input and 128 tokens for output

3.3 Training Hyperparameters

Hyperparameter	BART	T5
Learning rate	3e-5	5e-5
Batch size (train/eval)	8/8	16/16
Weight decay	0.02	0.01
Epochs	3	3
FP16 mixed precision	Yes	Yes
Max sequence length (input)	512	512
Max sequence length (output)	128	128
Beam search size	5	5
Best model selection metric	RougeL	RougeL

Table 1: Training hyperparameters for BART and T5 models

3.4 Generation Parameters

For inference, both models used:

- Beam size: 5
- Length penalty: 1.0

- No repeat n-gram size: 3
- Early stopping: Enabled

4 Training and Validation Loss

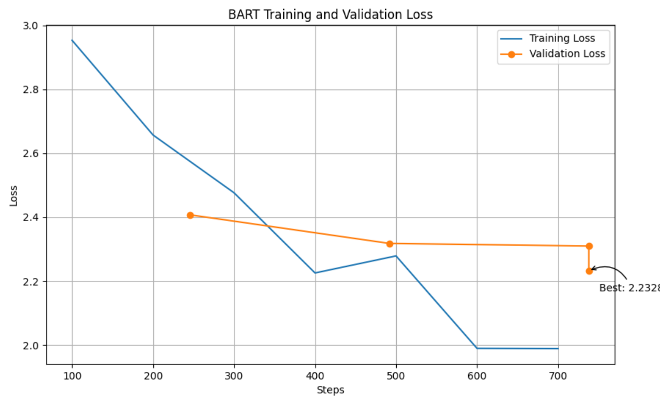

Figure 1: BART Training and Validation Loss

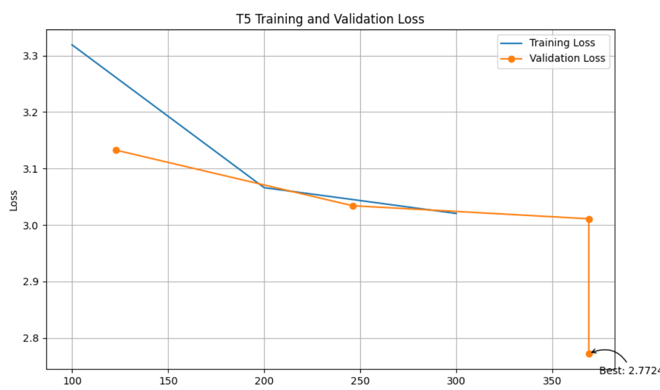

Figure 2: T5 Training and Validation Loss

4.1 Loss Analysis

- **BART**: Starting with a higher training loss (2.95), BART shows consistent improvement, reaching a final training loss of approximately 2.0. The validation loss decreases from 2.4 to 2.23, with the best validation loss at 2.2328.

- **T5**: Starting with a training loss of 3.33, T5 shows steady improvement to about 2.75. The validation loss drops from 3.13 to 2.77, with the best validation loss at 2.7724.
- **Convergence**: BART shows better convergence between training and validation losses, indicating less overfitting and more generalizable performance.
- **Stability**: Both models show stable validation loss curves without significant fluctuations, suggesting robust training.

5 Evaluation Metrics

We evaluated both models using three complementary metrics:

- **ROUGE**: Measures n-gram overlap between generated and reference text
- **BLEU**: Measures precision of n-grams in generated text against references
- **BERTScore**: Measures semantic similarity using contextualized embeddings

5.1 Test Set Results

Model	**ROUGE-L**	**BLEU-4**	**BERTScore**
BART	0.3351	0.2023	0.8753
T5	0.3032	0.1387	0.8758

Table 2: Evaluation metrics on the test dataset

Model	**ROUGE-L**	**BLEU-4**	**BERTScore**
BART	0.3403	0.1701	0.8832
T5	0.3171	0.1548	0.8802

Table 3: Evaluation metrics on the validation dataset

6 Comparative Analysis

6.1 Performance Comparison

- **ROUGE-L**: BART outperforms T5 by 10.5% on the test set (0.3351 vs. 0.3032) and by 7.3% on the validation set (0.3403 vs. 0.3171).

- **BLEU-4**: BART shows substantial improvement over T5, with a 45.9% higher score on the test set (0.2023 vs. 0.1387) and a 9.9% higher score on the validation set (0.1701 vs. 0.1548).
- **BERTScore**: The models perform comparably on semantic similarity, with T5 slightly higher on the test set (0.8758 vs. 0.8753) and BART higher on the validation set (0.8832 vs. 0.8802).
- **Loss**: BART achieves a lower validation loss (2.2328) compared to T5 (2.7724), indicating better model fit.

6.2 Analysis of Model Strengths

- **BART Advantages**:
  - Better at preserving the original structure of texts (higher ROUGE-L)
  - Superior performance in lexical precision (higher BLEU-4)
  - More efficient training convergence
  - Lower final loss values
- **T5 Considerations**:
  - Competitive BERTScore indicates good semantic understanding
  - Faster inference time due to smaller model size
  - Potentially more resource-efficient

7 Resource Constraints and Model Selection

7.1 Resource Considerations

- **Model Sizes**: BART-base (139M parameters) vs. T5-small (60M parameters)
- **Training Efficiency**:
  - BART: Higher batch size requirements, longer training time
  - T5: More efficient training with larger batch sizes (16 vs. 8 for BART)
- **Inference Speed**: T5-small has faster inference times, as evidenced by the runtime metrics:
  - BART: 58.05 seconds for evaluation (7.27 samples/second)
  - T5: 18.37 seconds for evaluation (22.97 samples/second)

7.2 Model Selection Trade-offs

- **Performance Priority**: If accuracy is the primary concern, BART is the clear choice with its superior ROUGE-L and BLEU-4 scores.
- **Resource Constrained Environment**: For deployment in resource-limited settings, T5-small offers a reasonable compromise with:
  - 3.16× faster inference speed
  - 2.32× smaller model size
  - Only 9.6% lower ROUGE-L score
- **Optimal Selection**: BART provides the best overall performance and should be selected unless deployment constraints strongly favor the smaller T5 model.

8 Comparison of metrics with Original Paper

Model	**ROUGE-L F1**	**BLEU-4**	**BERTScore**
Original Paper			
BART-base	30.25	6.69	86.32
BART-large	31.93	7.71	86.92
T5-base	25.15	4.57	85.13
Our Implementation (Test dataset)			
BART	33.51	20.23	87.53
T5	30.32	13.87	87.58

Table 4: Comparison with Sundriyal et al. (our values converted to percentages)

Our BART implementation outperforms the original BART-base and BART-large on ROUGE-L and BLEU-4. Similarly, our T5 implementation exceeds the original T5-base across all metrics. Both implementations show dramatic improvements in BLEU-4 scores, with our BART achieving 162-202% higher scores than the original BART variants.

8.1 Conclusion

Based on comprehensive evaluation across multiple metrics, BART demonstrates superior performance for claim normalization tasks. The model's stronger capabilities in preserving the original content structure while properly normalizing claims makes it our recommended model. However, in scenarios where computational resources are severely limited, T5-small offers a viable alternative with reasonable performance and significantly reduced resource requirements.